

IPL DATA ANALYSIS DASHBOARD USING POWER BI

1. Problem Statement

The objective of this project is to analyze Indian Premier League (IPL) match and delivery data using Power BI to uncover meaningful insights related to team performance, player statistics, match outcomes, and game trends. The project focuses on transforming raw cricket data into interactive dashboards and analytical insights for better decision-making.

2. Business Use Cases

- Identifying the most successful teams and players in the IPL.
- Determining the factors that contribute to a team's or player's success.
- Analyzing the impact of different playing conditions on the outcome of matches.
- Identifying trends in IPL data over time.
- Generating insights that can be used to improve the performance of IPL teams and players.
- Player selection and auction strategies.
- Performance analysis and optimization.
- Augmented coaching and data-driven analysis.
- Enhancing fan engagement.
- Improving broadcasting strategies

3. Dataset Overview

The dataset used in this project contains historical Indian Premier League (IPL) match data and ball-by-ball delivery data. The data was used to perform team analysis, player performance evaluation, and match trend analysis using Power BI.

Two datasets were used in this project:

Table Name	Description	Dataset Size
Matches	Contains match-level information such as season, venue, teams, toss details, match results, and player of the match	636
Deliveries	Contains ball-by-ball delivery information including batsman, bowler, runs scored, extras, wickets, and innings details	150,460

Matches Table

- Match ID
- Season
- City
- Venue
- Toss Winner
- Toss Decision
- Match Winner
- Player of the Match

Deliveries Table

- Match ID
- Innings
- Over and Ball Number
- Batsman
- Bowler
- Total Runs
- Extras
- Wicket Information

4. Data Cleaning Process

Matches Table

Issue Identified	Action Taken	Reason
Null values in the city column	Replaced using corresponding venue information	Venue details were available, allowing recovery of missing city values accurately
Null values in player_of_match	Replaced with "No Player of Match"	These matches had no official player award due to match abandonment or no result
Missing umpire names	Replaced with "Not Available"	Improved readability and prevented blank values in dashboard visuals
umpire3 column contained all null values	Removed the column	The column had no analytical or business value
Null values in winner column	Replaced with "No Winner"	Some matches were marked as "No Result," meaning no official winner was declared

These cleaning steps improved data quality and ensured better reporting and dashboard usability in Power BI.

Deliveries Table

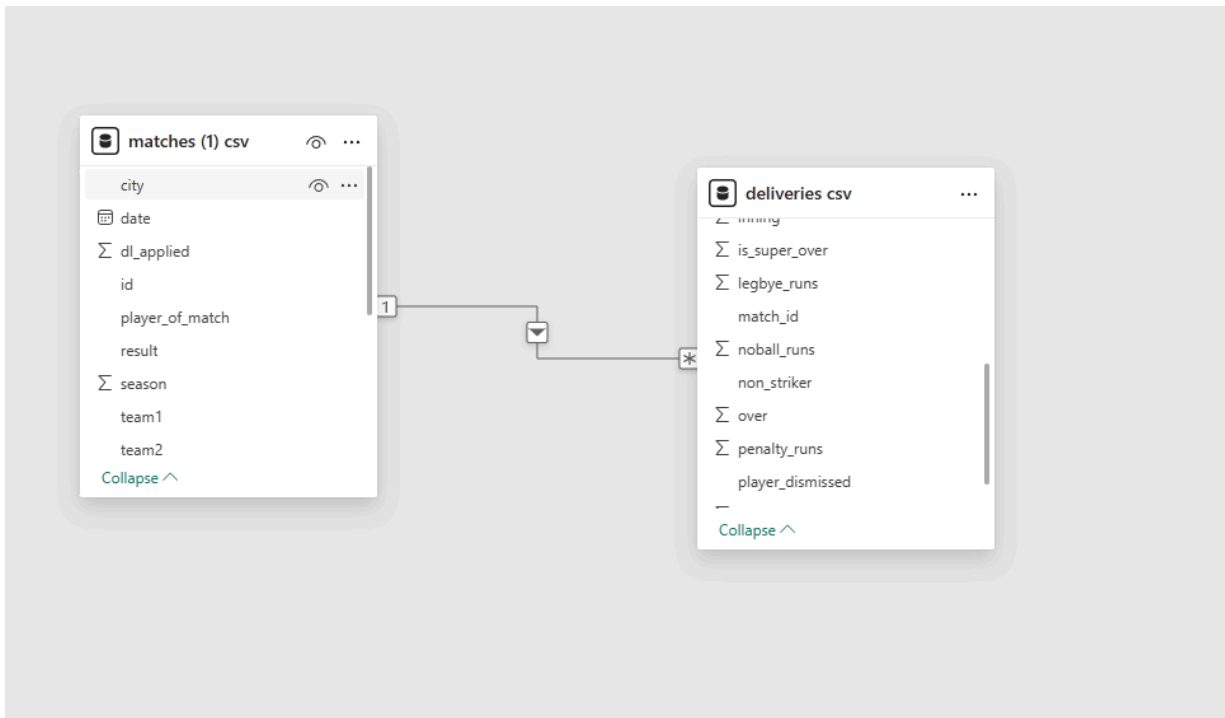
Data cleaning and transformation steps were performed on the Deliveries table

Issue Identified	Action Taken	Reason
Null values in player_dismiss	Replaced with “Not Out”	Most deliveries do not result in wickets
Null values in dismissal_kind	Replaced with “No Dismissal”	Indicates deliveries without dismissals
Null values in fielder	Replaced with “No Fielder Involved”	Many wickets do not involve fielders
Duplicate delivery check	Created Delivery_ID and verified uniqueness	Ensured delivery-level data integrity
Team naming inconsistencies	Standardized team names	Prevented fragmented analysis across visuals

These transformations improved data quality, enhanced dashboard readability, and ensured accurate analytical reporting.

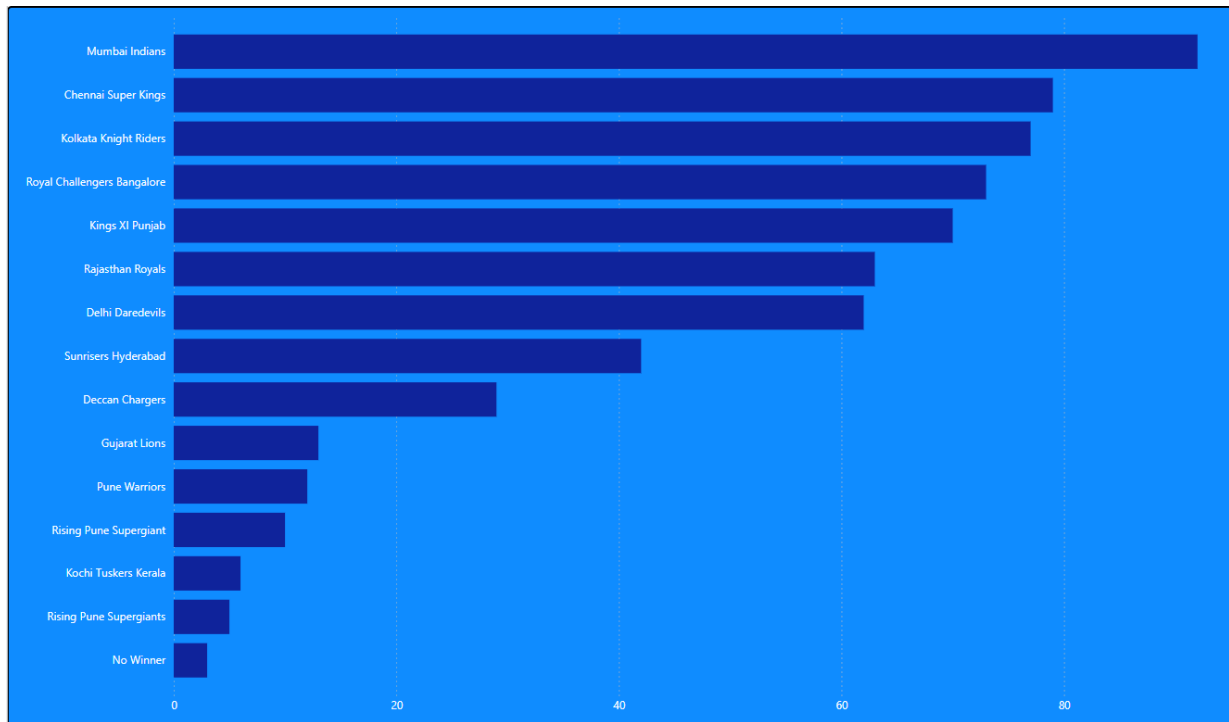
5. Data Modeling

Created Relation in both tables by making relationships in id columns from matches table to Match_id column from Deliveries table .



6. Important Visuals

1.To identify the most successful IPL teams based on total match victories across all seasons.



- Mumbai Indians recorded the highest number of match wins among all IPL franchises.
- Chennai Super Kings and Kolkata Knight Riders also demonstrated strong overall performance.
- Teams with stable leadership and balanced team composition maintained consistent success over multiple seasons.

2. To analyze the top-performing batsmen based on total runs scored in IPL history.



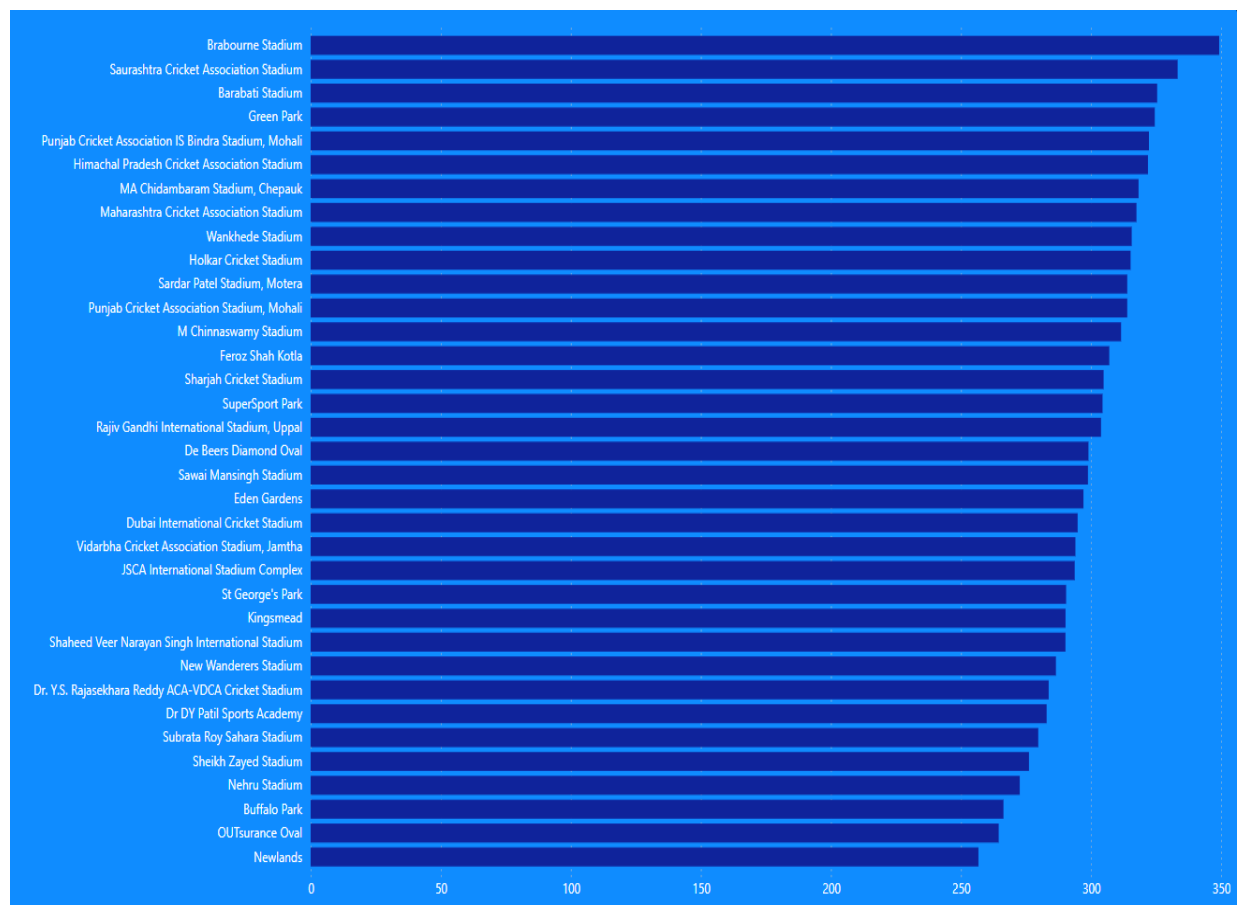
- SK Raina and Virat Kohli consistently ranked among the highest run scorers.
- It states That An Experienced player perform better in throughout multiple season
- Teams with reliable top-order batsmen generally achieved better match stability
- Team should focus on Picking Good experience player which can perform and lead young players.

3. To identify the most successful bowlers based on wicket-taking performance.



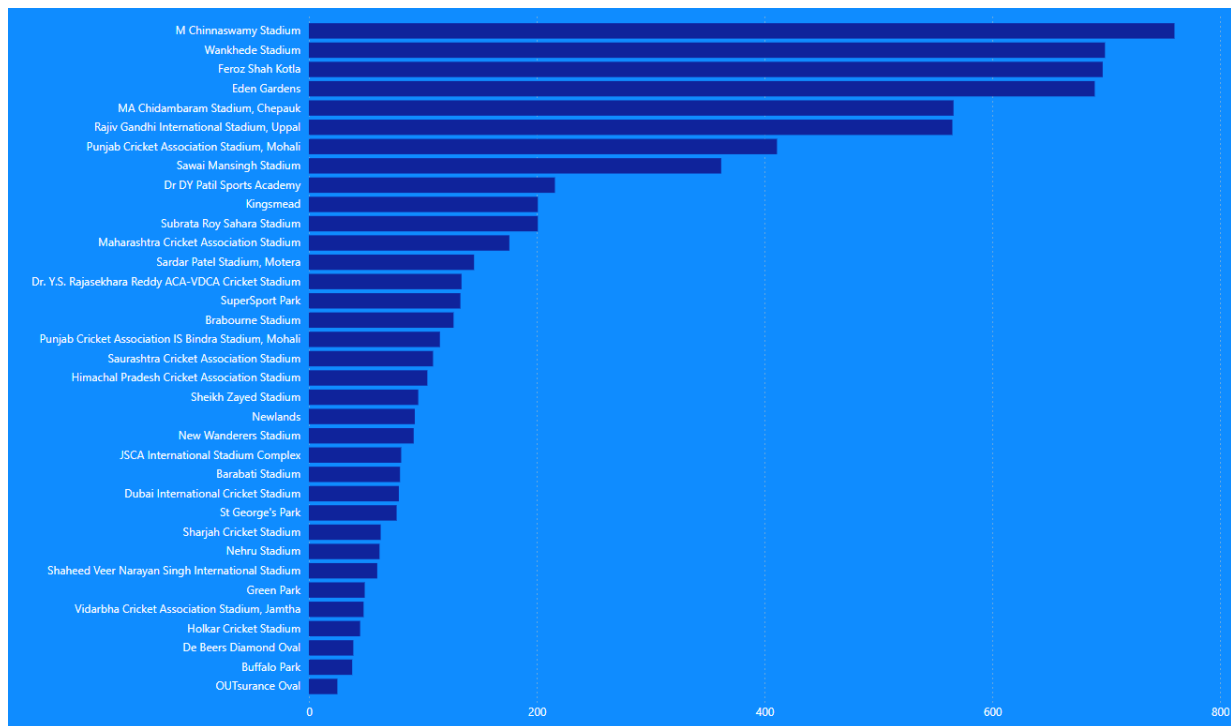
- SL Malinga emerged as one of the leading wicket-taking bowlers in IPL history.
- Bowlers with strong death-over execution and consistency significantly impacted match outcomes.
- Teams with effective bowling attacks often maintained better defensive performance during high-pressure matches.

4. To identify batting-friendly venues based on scoring trends and average runs.



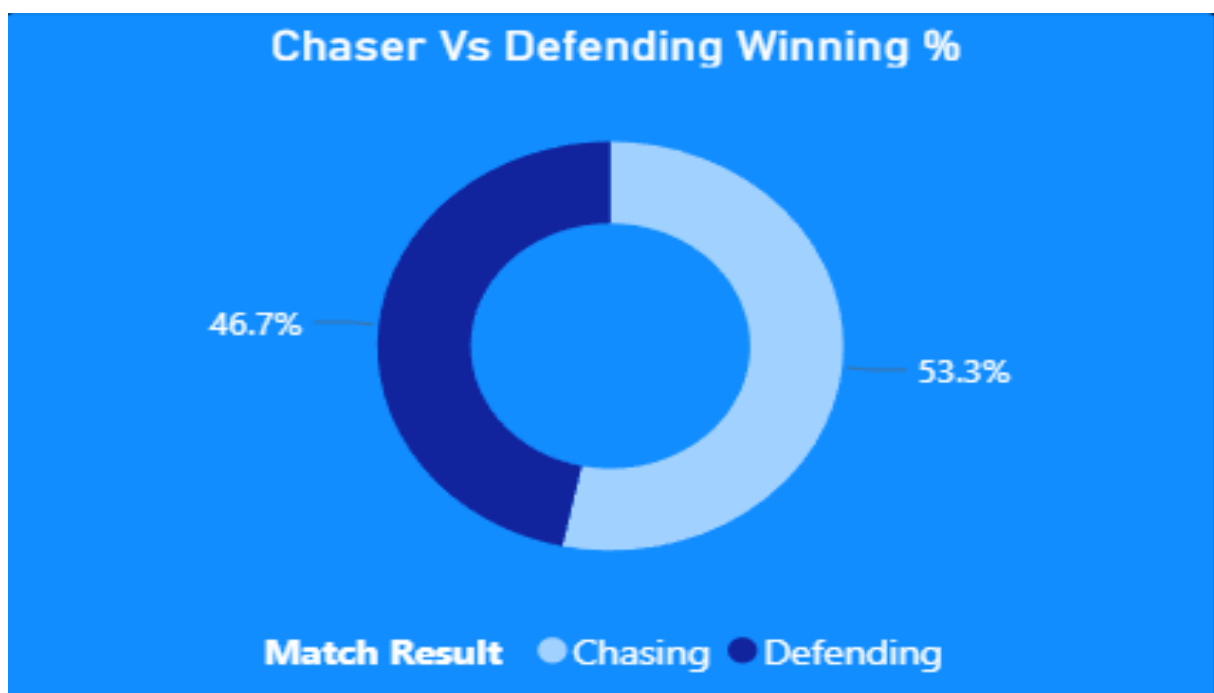
- Certain venues consistently produced high-scoring matches like Brabourne stadium and Saurashtra cricket association stadium.
- Flat pitches and shorter boundaries contributed to batter-friendly conditions.
- Batting performance varied significantly depending on venue characteristics.

5. To identify bowling-friendly venues based on wicket frequency and scoring patterns.



- Some venues supported bowlers through lower average scores and higher wicket counts like m chinnaswammy stadium and wankhede stadium.
- Pitch conditions played a major role in bowling performance like rough pitches helps spinners while green pitches helps fast bowlers .
- Bowling-friendly stadiums often produced more competitive and low-scoring matches which eventually benefit the chasing team.

6. To analyze whether teams were more successful while chasing targets or defending totals in IPL matches.



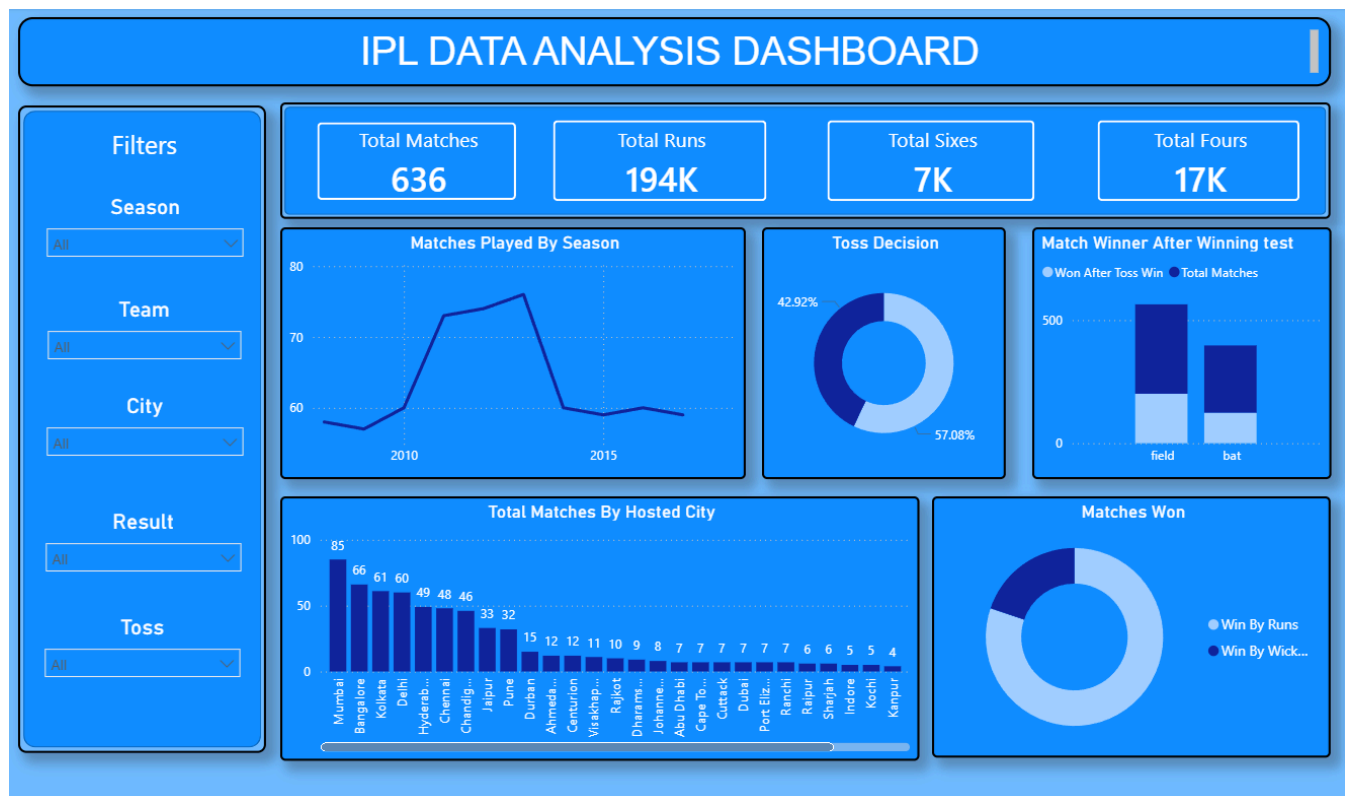
- Teams chasing targets won approximately 53.3% of matches, while defending teams won around 46.7%.
- The data indicates a slight advantage for teams batting second.

- Factors such as dew conditions, pitch behavior, and target visibility may have contributed to higher chasing success.

7. Business Recommendation

1. The analysis showed that teams with stable and consistent performers achieved better long-term success across IPL seasons. Franchises should focus on retaining reliable batsmen, bowlers, and all-rounders instead of making exclusive team changes every season.
2. Venue analysis revealed that certain stadiums strongly favor either batting or bowling performances. Teams should adapt their playing XI, bowling combinations, and batting approach according to venue conditions to maximize performance.
3. The dashboard analysis indicated that chasing teams had a slightly higher win percentage compared to defending teams. Teams should improve middle-order stability, finishing ability, and pressure-handling skills for successful run chases.
4. Toss analysis showed that teams frequently preferred fielding first due to changing match conditions and chasing advantages. Teams can use historical venue data and toss trends to make more effective toss-related strategic decisions.
5. Bowling analysis demonstrated that successful teams often had strong wicket-taking bowlers and effective death-over bowlers. Teams should focus on recruiting bowlers capable of controlling runs and taking wickets under high-pressure situations.

8. Dashboard



TEAM PERFORMANCE ANALYSIS

Filters

Season

All

Team

All

City

All

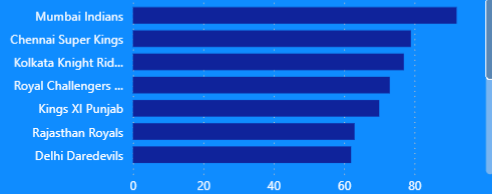
Most Successful Team
Mumbai Indians

Total Teams
14

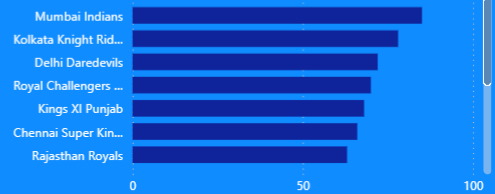
Total Sixes
7K

Total Fours
17K

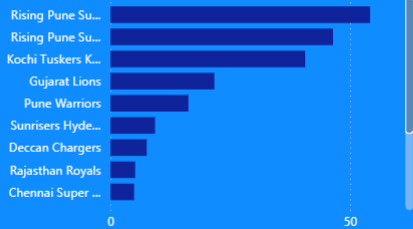
Matches Win By Teams



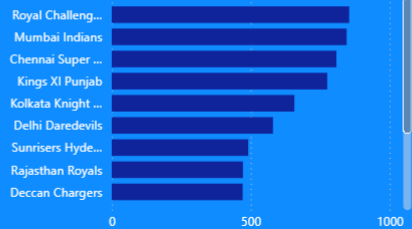
Toss Win By Teams



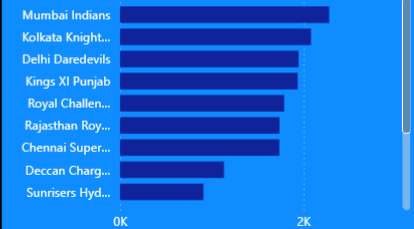
Win Percentage After Winning Toss



Sixes By Teams



Fours By Teams



PLAYER PERFORMANCE ANALYSIS

Filters

Season

All

Team

All

Player

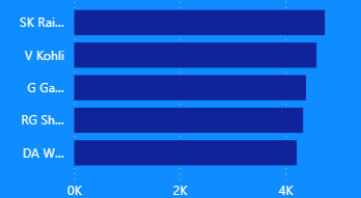
All

Top Batter
SK Raina

Top Bowler
SL Malinga

Top Fielder
KD Karthik

Best Batsman



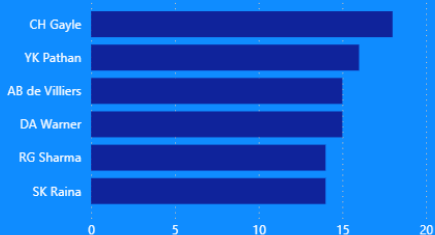
Best Bowlers



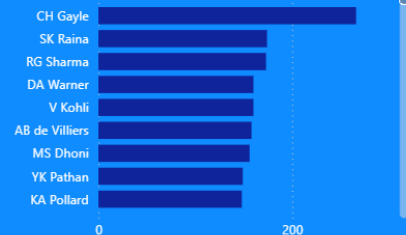
Best Fielders



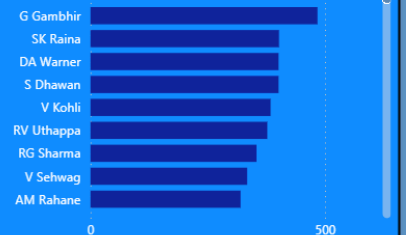
MOST PLAYER OF THE MATCH AWARDS



Most Sixes By Player



Most Fours By Players



VENUE & MATCH CONDITION ANALYSIS

Filters

Season

All

Team

All

Venue

All

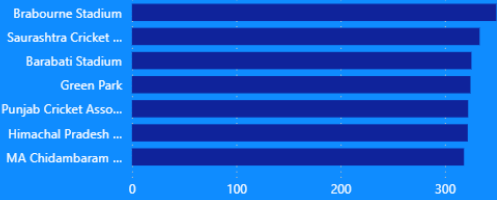
Total Venues

35

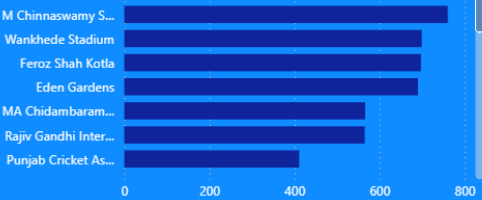
Venue Highest Matches

Rajiv Gandhi International Stadium, ...

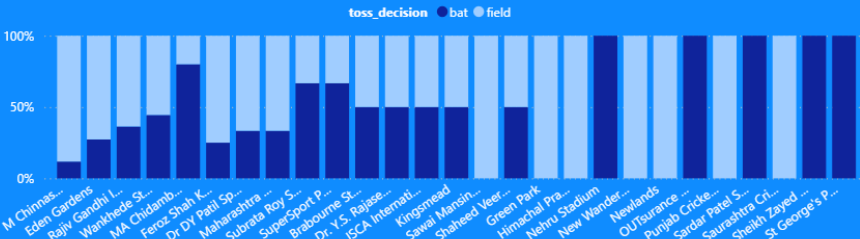
Best Venue For Batters



Best Venue For Bowlers



Toss Decision By Venue



Most Sixes By Player

